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Smart Car Parking System

ABSTRACT

Due to traffic and parking problems in the present urban areas, safe and easy parking system is very important for drivers. Parking management is a major challenge, especially in residential and small business areas.The purpose of this study was to design and implement a smart car parking system, which could be used at low cost and easily. The system will automatically complete the car parking process and will be convenient for users. The study uses Arduino microcontroller, infrared (IR) sensor and cervo motor. When the car enters, the gate opens automatically and the gate is closed when it comes out. The condition of the parking slot is displayed in real-time using the LCD display. The system is able to successfully detect the vehicle entry and exit and the gate automatically controls. LCD display helps users to easils understand the condition of the parking slot. This smart parking system is low cost effective and convenient for users. This will help facilitate parking management in residential and small business areas. In the future, the system can be further enhanced commercially.

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INTRODUCTION

The number of vehicles in the present urban areas is increasing rapidly, resulting in a serious challenge of parking problems. Studies show that about 5% of the city's traffic drivers drive to find parking, which causes time and fuel and increases environmental pollution. To solve this problem, the need for smart parking systems is increasing day by day.

The parking process can be automatically and efficiently using modern technology such as Arduino, Infrared (IR) sensors, Servo Motors and LCD display in the smart parking system. Using these technologies the entrance and exit of the vehicle can be detected and the parking slot condition can be displayed in realtime.

In this study we have designed a low cost, easy -to -implement smart parking system, which is suitable for small businesses and residential areas. this will make the parking process easier, faster and safe for drivers.

Parking problems in urban areas are a common phenomenon, which causes time and fuel for drivers and enhances environmental pollution. Studies show that about 5% of the city's traffic drivers drive to find parking, which causes time and fuel and increases environmental pollution.

The current parking systems are often manual, which are time -consuming and inefficient for drivers. There is a need for smart parking systems to solve this problem, which can automatically detect the vehicle entry and exit and show the condition of the parking slot in realtime.in this study we have designed a smart parking system, which uses Arduino, IR sensor, pervo motor and LCD display. The system detects the access and exit of the vehicle automatically opens and shuts the gate and the parking slot is displayed on the LCD display. This system is low cost, easily implementable and suitable for small businesses and residential areas. This will make the parking process easier, faster and safe for drivers.

Design and implement an automated smart parking system. Used IR sensors to detect vehicle entry and exit. Use of servo motor to control the gate. The condition of the parking slot is used to display the LCD display to display in real-time. Creating a low cost, easy implementable system, which is suitable for small businesses and esidential areas.

A functional smart parking system has been designed and implemented. The system automatically controls the gate by detecting the vehicle entry and exit. The condition of the parking slot is displayed on the LCD display in real-time. He system is low cost, easily implementable and suitable for small businesses and residential areas. This research has made an important contribution to the development of the smart parking system.

Motivation and related works: (ID:232-35-571)

Now a days the number of cars increasing day by day and for those car needs a lot of parking space. But this manual system of parking is so time consuming and for this car drivers often park their car in the road and creates chaos. For this we came up with a solution, using robotics er are creating smart parking system, where everything will be automatic, no need any human interference.

Recently there are some projects and paper has been published related to our project. After research those we found them very useful.

1. **Smart Parking System Using IoT and Image Processing (2019)**
   Method: Used ultrasonic sensor and image recognition to find out empty parking slot.
   Result: peoples are using parking more than before.
   Contribution: Real time monitoring.
   Limitation: In low light its not that much useful.

2. **RFID-Based Smart Car Parking System (2018)**
   Method: they attached a RFID card to detect the car.
   Result: It became more systematically than before.
   Contribution: It provide safety and authentication.

   Limitation: Without attaching RFID card, the system can't detect.

3. **Automated Parking Using Arduino and IR Sensors (2020)**
   Method: IR sensor detect the movement and opens gate using servo motor.
   Result: Provides automation.
   Contribution: Less expensive and easy to setup.
   Limitation: In sun light its not works perfectly.

4. **License Plate Recognition Based Entry System (2021)**
   Method: Used number plate recognition system using camera.
   Result: In day light it provides 88% accuracy.
   Contribution: Fully automatic.
   Limitation: In night time, rainy day and foggy day it can't read the number plate.

5. **Smart Car Parking System Using GSM and Arduino (2020)**
   Method: Sends SMS to the user if there any parking slots are available.
   Result: User don't have to wait long time.
   Contribution: Provides monitoring system.
   Limitation: Depend on GSM network.

6. **IoT-Based Real-Time Smart Parking System (2022)**
   Method: Made an apps connected with the NodeMCU device which can read the empty space and show it on apps.
   Result: Users can reserve parking slot.
   Contribution: Scalable and interactive system.
   Limitation: Expensive setup.

7. **Automatic Car Gate System Using PIR Sensors (2019)**
   Method: PIR sensor detects movement and opens gate for car.
   Result: Gates can open automatically.
   Contribution: Detect movement accurately.
   Limitation: It detects human and animal's movement too.

8. **Smart Parking Using Mobile App and Cloud Integration (2021)**
   Method: System keeps parking record and send it to the user.
   Result: User can see real time observation.
   Contribution: User friendly and easy.
   Limitation: If server become down the system will crash.

9. **AI-Based Vehicle Detection for Parking Automation (2023)**
   Method: Used YOLO model to detect car.
   Result: Accuracy up to 93%.
   Contribution: Included Deep Learning Technology into the system.
   Limitation: Costly because the system needs GPU for real time working.

10. **Solar-Powered Smart Parking System (2020)**
    Method: Integrate solar panel and smart sensor for parking system.
    Result: In sunlight it works perfectly.
    Contribution: Eco friendly.
    Limitation: Without proper sunlight it wont works perfectly.

Identification of Research Gaps

- Maximum time the automatic gate can't open properly.
- Not everyone implants display systems.
- Real time detection is not useful that much.
- In small space those project works properly but when it gets bigger space it can't work properly.
- They don't have enough backup plan for the system.

- Didn’t built user friendly interface for users.
- They didn’t implement all those sensors and device in their system.
- In low light the system didn’t work properly.

We are trying our best to overcome these and build a perfect Smart Car Parking System. It will help to build smart city and solve real parking problem.

Methodology

In this project er are making Smart Parking System, er are using IR sensor for car detection to open and close the door automatically. In this process we are using a display to show the message.

Our system promised that there is no need of human interference in parking system.

Components

- IR sensor (x2)
- Arduino UNO
- Servo Motor
- LCD Display

Theoretical Framework

IR sensor: IR sensor emits infrared light, if it gets block by any objects, it gets bounce back to the sensor and send signal to the Arduino. That's how it detects car.

Arduino UNO: Arduino is a mini computer. It can control other devices. In our project it takes IR sensors input and send it to the Servo motor as output. And display shows it.

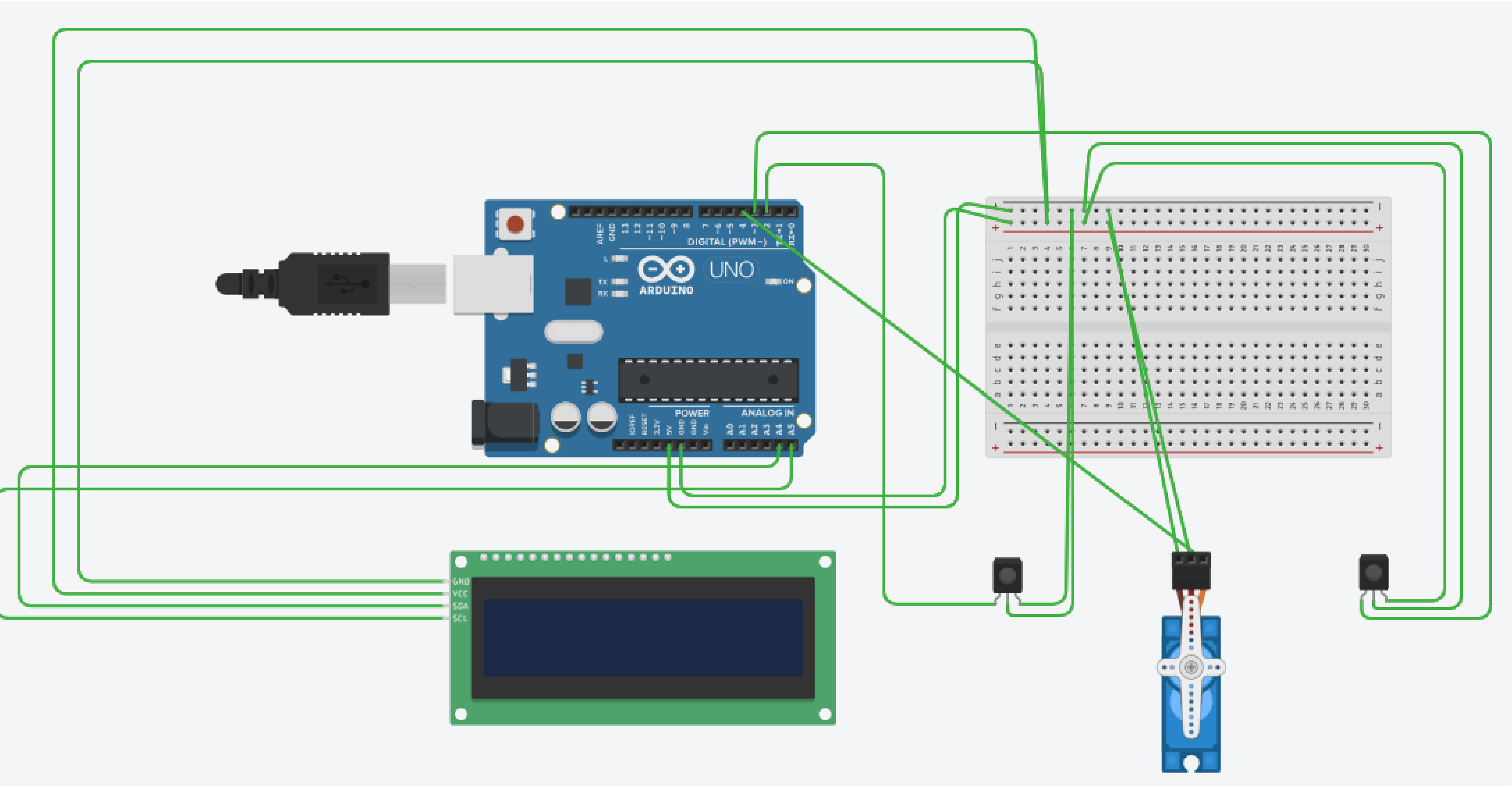
Servo Motor: We are using this to opens and closes gate.

LCD Display: we are using 16x2 LCD Display to show messages of gates.

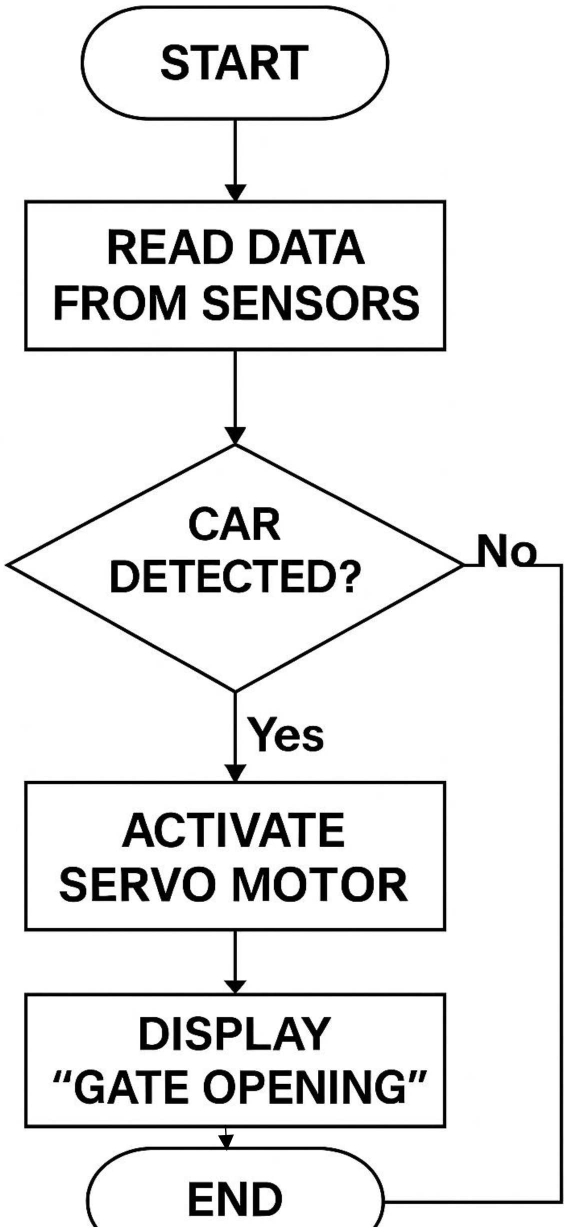
Automation System: Our system provides automation where humans interference is not necessary to connects all those devices.

Hardware Design and Software Flowchart

Hardware Design:



Software Flowchart:



Every time the sensor detects any car, the program initiates to open and close the gate.

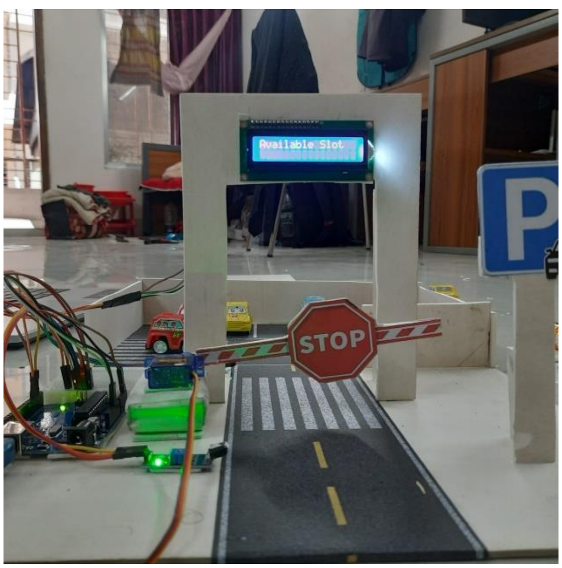
This project will help to build smart city and this will help to solve real parking problem.

RESULT AND DISCUSSION

1.PRESENTATION OF KEY FINDINGS:

The main purpose of our project "smart car parking system" was to create an automated parking system that could recognize the vehicle entry and departure and show the gate control and parking status accordingly. To meet this purpose we have used Arduino Uno, two IR sensors, a Servo motor and an LCD display. After the implementation of the entire system, we collect the results by experimenting the car multiple times. The tests were mainly divided into three sections the effectiveness of the sensor, the accuracy of the gate operation and the clear update of the display.

Some of the main results of the test are presented in the table below:



Front View

Upper View

2.INTERPRETATION OF RESULT:

It is clear from these results that the system has been able to works accurately. particularly, the status update through the LCD display has been successful in 5% of cases, which creates helpful experiences for the user. The result analysis shows that IR sensors have done good response when the car enters(without black and white car) because the vehicle's speed is slow and the sensor has been able to send timely signals. The signal is sometimes a little late when the car is out specially when the car is going out. As a result, the Response of the Servo Motor has been 2-3 seconds late.

The way we created the code logic, the gate has received the right opening and closing signal without human help and has shown immediate message on the display. Another aspect was that we were able to deliver appropriate messages (Gate Closed, Available Slot each time in both the car's entry and departure. It has served as a visual guide for the user and has made the system easier.in the case of gate operation, the servo motor is at some point slowly, which is probably due to the temporary variation of electricity of the motor. However, it has been done a lot by repeatedly examining it. These results prove that our system is effective and automatically managed for the small range parking system. It can become commercially useful through further development in the future.

3.LIMITATION OF THE STUDY:

Although the smart car parking system created by us has given some good results, some limitations have been noticed during the study. First, the IR sensor works well at a certain distance and angle if the car passes a little or the corner comes into the corner, then the sensor fails to detect occasionally. Second, the light of the environment varies, such as in the darkness of the sun or at night, the sensor response has shown some problems . Third, the Servo motor gets a little hot after a long period of time and then its speed is slightly reduced. we still have no cloud or app integration in this version - which provided users from far away. Another restriction is that our system has only worked with two sensors, so it needs to make a further scale up for large scale parking spaces and add advanced sensors and modules. Due to these limitations, this system is not very large in size now, but it is possible to make it stronger and realistic through further research and development in the future.

5. CONCLUSION

The Smart Car Parking System effectively demonstrates how Arduino technology can automate and streamline parking management. By using IR sensors to detect vehicle presence and a servo motor for gate control, the system simplifies the parking process and displays real-time slot availability via an LCD. This low-cost, user-friendly setup is ideal for homes and small businesses, offering an efficient alternative to manual parking systems. Key challenges like sensor calibration and component integration were resolved through testing and iteration. Future improvements may include mobile app integration, cloud-based monitoring, and solar power support, making the system more intelligent and scalable for larger applications. Overall, the project highlights the potential of embedded systems in developing smart, accessible solutions for everyday problems.

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